

Stage 1

Before starting with Stage 1 you must check the mode in which the wlan0 interface is for this the reserved command `vcom` will be used, it must be shown as in **Image 1**.

```
epn@raspberrypi:~ $ vcom
# interfaces(5) file used by ifup(8) and ifdown(8)
# Include files from /etc/network/interfaces.d:
source /etc/network/interfaces.d/*

allow-hotplug wlan0
#auto wlan0
iface wlan0 inet static
address 10.0.1.1
netmask 255.255.255.0
network 10.0.1.0
broadcast 10.0.1.255
epn@raspberrypi:~ $
```

Image 1. WLAN0 check.

Otherwise, the reserved command `com` should be used to change the interface mode to `hotplug`, as shown in **Figure 2**.

```
#allow-hotplug wlan0
auto wlan0
iface wlan0 inet static
address 10.0.1.1
netmask 255.255.255.0
network 10.0.1.0
broadcast 10.0.1.255
epn@raspberrypi:~ $ com
[sudo] password for epn:
Line auto wlan0 commented out on /etc/network/interfaces
epn@raspberrypi:~ $ vcom
# interfaces(5) file used by ifup(8) and ifdown(8)
# Include files from /etc/network/interfaces.d:
source /etc/network/interfaces.d/*

allow-hotplug wlan0
#auto wlan0
iface wlan0 inet static
address 10.0.1.1
netmask 255.255.255.0
network 10.0.1.0
broadcast 10.0.1.255
```

Image 2. WLAN0 mode switching.

Now we will proceed to configure the essential parameters for the operation of the network, we must use the reserved `hot` command that will open two terminals, the first with information related to the `hotplug`, it is recommended to modify only the field of `ssid` and `wpa_passphrase` so that it is related to the place where the demonstration is being carried out.

The second terminal will show the range that the network will have, it is configured to accept 98 hosts, this field can be modified according to the number of participants of the demonstration.

```
epn@raspberrypi:~ $ hot

#dhcp-name-match=set:wpad-ignore,wpad
#dhcp-ignore-names=tag:wpad-ignore

interface=wlan0
dhcp-range=10.0.1.2,10.0.1.100,255.255.255.0,24h

interface=wlan0
driver=nl80211
ssid=Demo
hw_mode=g
channel=6
wmm_enabled=0
macaddr_acl=0
auth_algs=1
ignore_broadcast_ssid=0
wpa=2
wpa_passphrase=12345678
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP
```

Image 3. Hot settings.

To start Stage 1 you must use the reserved command `ss1` or `SS1`, which will turn on the services necessary for the wlan0 interface to work as a hotplug, the system will automatically restart once this process is finished.

```
epn@raspberrypi:~ $ ss1
Synchronizing state of hostapd.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable hostapd
Synchronizing state of dnsmasq.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable dnsmasq
```

Image 4. Stage start.

Once the system restarts, there will be a change of icon in the taskbar that will confirm the change of mode, it is important to note that from that moment on the previously configured network will become visible, and all Wi-Fi connection functionality will be lost, including that of viewing available networks, so to have an internet connection it must be provided via ethernet.

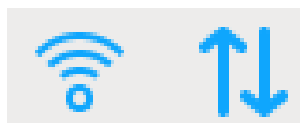


Image 5. Interface mode icon.

Next, we will explain the method to share internet to the device from a Windows machine, you can connect the device directly to an AP if available and skip this section.

The Network Connections window must be opened from the Control panel, here the interface of the machine that is receiving internet must be selected and its properties opened, previously the device and the machine in question must have been connected via ethernet.

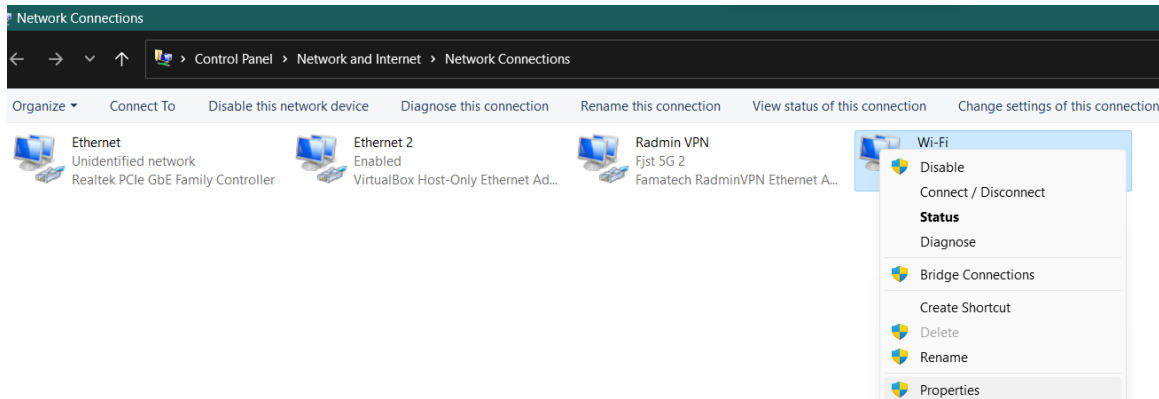


Imagen 6. Network Connections.

In the properties, you must go to the Sharing tab in which you must activate the box "Allow other network users to connect through this computer's Internet connection", and select the ethernet option in the "Home networking connection" section.

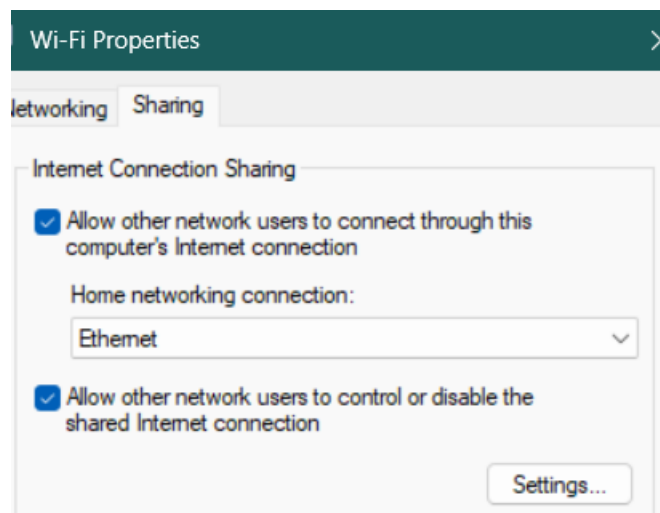


Imagen 7. Internet Connection Sharing.

Once this process is finished, the internet will be available on the device, and by extension the connections offered by the hotplug will also have access, as shown in **Image 8**. Internet service is available.

```
epn@raspberrypi:~ $ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=21.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=20.4 ms
^C
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 20.442/20.729/21.017/0.287 ms
```

Image 8. Internet Sharing Check.

In **Image 9**. An Android device's connection to the Promedex device's network is displayed.

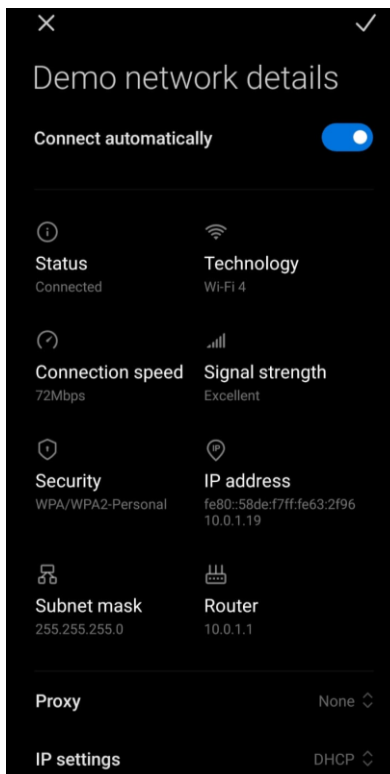


Image 9. Grid connection.

At this point, participants will be asked to enter the network to interact with the stage, the first thing that will be made known to them, in addition to not entering unknown networks that are not protected with a password and that in many cases offer free internet, is how their electronic devices can be identified, for this they will use two reserved commands, the `con` and the `fcon`.

```
epn@raspberrypi:~ $ con
Address HWtype HWaddress Flags Mask Iface
LAPTOP-17BJFRV9.mshome. ether 98:fa:9b:5c:dd:4a C eth0
Xiaomi-11-Lite-5G-NE ether 5a:de:f7:63:2f:96 C wlan0
epn@raspberrypi:~ $ fcon
LAPTOP-17BJFRV9.mshome.net (192.168.137.1) at 98:fa:9b:5c:dd:4a [ether] on eth0
Xiaomi-11-Lite-5G-NE (10.0.1.19) at 5a:de:f7:63:2f:96 [ether] on wlan0
```

Image 10. with and fcon.

Continuing with the demonstration, the exhibitor must carefully explain that access to any page that the participants wish will be allowed, for a limited time, the timed traffic capture will be initiated with the reserved command `ts` or `TS`, a message will be displayed in which it is specified that the time of the capture must be entered manually in seconds, Once the capture is complete, a message and the number of packets captured so far will be displayed.

```
epn@raspberrypi:~ $ ts
Enter time in seconds
60
[sudo] password for epn:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'wlan0'
6468 Shark is hungry Yum-yum!
```

Image 11. Packet capture.

Once the capture process is finished, the exhibitor must explain to the participants that all the information we send or receive is protected by the companies to avoid leaks of sensitive information, and the reserved command `lk` will be executed that will allow the participants to observe the illegible traffic of the capture, it can be explained that a similar process occurs in the transmission of messages by WhatsApp which is probably the most used application in the Ecuador for sending messages.

```
epn@raspberrypi:~ $ lk
[...]
```

Image 12. Unreadable capture.

Finally, filters will be applied to the capture to show specific information and in clear text, evidencing the power that attackers have to visualize the information of their victims, for this the reserved command `f1` will be used, which will execute a filter that will show the general information of the capture made.

```
epn@raspberrypi:~ $ f1
2085 2.749519350 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=734592 Win=2470 Len=0 TSval=2550138394 TSecr=2110276703
2086 2.746688868 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=755468 Ack=7680 Win=16454 Len=1228 TSval=2110276714 TSecr=2550138196 [TCP segment of a
assembled PDU]
2087 2.746698368 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=756696 Ack=7680 Win=16454 Len=1228 TSval=2110276714 TSecr=2550138196 [TCP segment
a reassembled PDU]
2088 2.747658835 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=737048 Win=2470 Len=0 TSval=2550138395 TSecr=2110276706
2089 2.747975345 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=739504 Win=2470 Len=0 TSval=2550138395 TSecr=2110276707
2090 2.748041022 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=741960 Win=2470 Len=0 TSval=2550138396 TSecr=2110276708
2091 2.748102376 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=744416 Win=2470 Len=0 TSval=2550138396 TSecr=2110276709
2092 2.748141751 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=746872 Win=2470 Len=0 TSval=2550138396 TSecr=2110276710
2093 2.748178105 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=749328 Win=2470 Len=0 TSval=2550138397 TSecr=2110276711
2094 2.748710292 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=757924 Ack=7680 Win=16454 Len=1228 TSval=2110276715 TSecr=2550138196 [TCP segment of a
assembled PDU]
2095 2.748728834 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=759152 Ack=7680 Win=16454 Len=1228 TSval=2110276715 TSecr=2550138196 [TCP segment
a reassembled PDU]
2096 2.748762063 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=760380 Ack=7680 Win=16454 Len=1228 TSval=2110276716 TSecr=2550138196 [TCP segment of a
assembled PDU]
2097 2.748776959 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=761608 Ack=7680 Win=16454 Len=1228 TSval=2110276716 TSecr=2550138196 [TCP segment
a reassembled PDU]
2098 2.748919927 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=751784 Win=2470 Len=0 TSval=2550138397 TSecr=2110276712
2099 2.749107114 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=754240 Win=2470 Len=0 TSval=2550138397 TSecr=2110276713
2100 2.7495906124 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=762836 Ack=7680 Win=16454 Len=1228 TSval=2110276717 TSecr=2550138196 [TCP segment of a
assembled PDU]
2101 2.749519041 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=764064 Ack=7680 Win=16454 Len=1228 TSval=2110276717 TSecr=2550138196 [TCP segment
a reassembled PDU]
2102 2.750776123 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=765292 Ack=7680 Win=16454 Len=1228 TSval=2110276718 TSecr=2550138196 [TCP segment of a
assembled PDU]
2103 2.750793154 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=766520 Ack=7680 Win=16454 Len=1228 TSval=2110276718 TSecr=2550138196 [TCP segment
a reassembled PDU]
2104 2.750847425 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [ACK] Seq=767748 Ack=7680 Win=16454 Len=1228 TSval=2110276719 TSecr=2550138196 [TCP segment of a
assembled PDU]
2105 2.750862529 149.154.166.120 - 10.0.1.19 TCP 1294 443 - 47828 [PSH, ACK] Seq=768976 Ack=7680 Win=16454 Len=1228 TSval=2110276719 TSecr=2550138196 [TCP segment
a reassembled PDU]
2106 2.751434924 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7680 Ack=756696 Win=2470 Len=0 TSval=2550138398 TSecr=2110276713
2107 2.751664090 10.0.1.19 - 149.154.166.120 TCP 315 47828 - 443 [PSH, ACK] Seq=7680 Ack=757924 Win=2470 Len=249 TSval=2550138400 TSecr=2110276714 [TCP segment of
reassembled PDU]
2108 2.751729871 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7929 Ack=760380 Win=2470 Len=0 TSval=2550138400 TSecr=2110276715
2109 2.752643047 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7929 Ack=762836 Win=2470 Len=0 TSval=2550138401 TSecr=2110276716
2110 2.752849609 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7929 Ack=765292 Win=2470 Len=0 TSval=2550138402 TSecr=2110276717
2111 2.753010443 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7929 Ack=767748 Win=2470 Len=0 TSval=2550138402 TSecr=2110276718
2112 2.753189036 10.0.1.19 - 149.154.166.120 TCP 66 47828 - 443 [ACK] Seq=7929 Ack=770204 Win=2470 Len=0 TSval=2550138403 TSecr=2110276719
2113 2.753463307 149.154.166.120 - 10.0.1.19 SSLv2 1204 Encrypted Data [TCP segment of a reassembled PDU]
```

Image 13. Filter capture.

Warning: End of Stage 1, Stage 2 is sequential and requires the pre-established connection in this Stage.

Prepared by JS.